

UEL DEMO

Cohesive Zone Modelling

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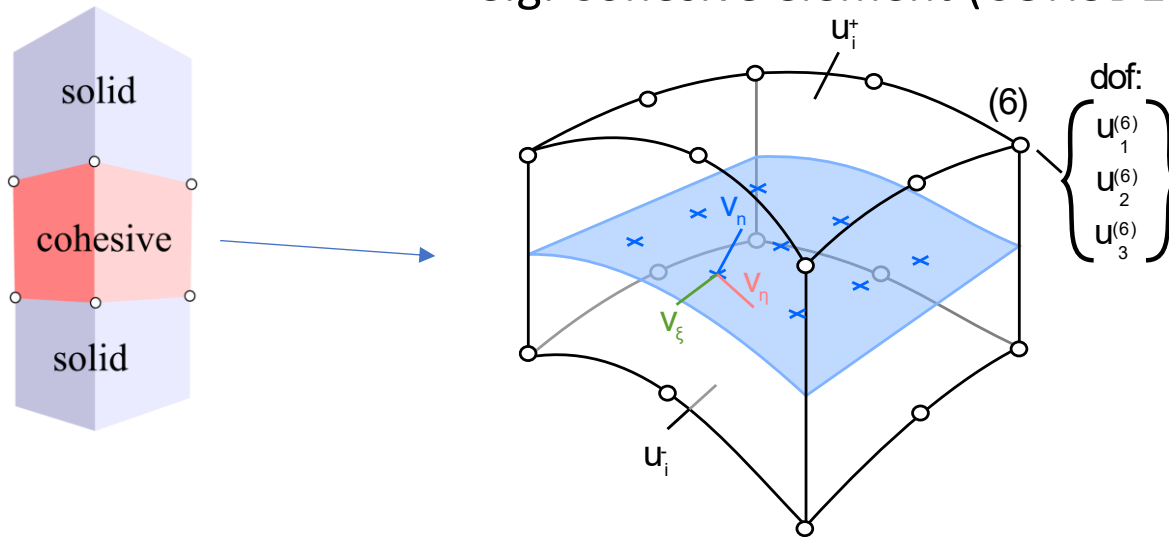
24.05.2026

FEM Framework

Opening displacements:

$$\Delta_i = \Theta u_i$$

e.g. Cohesive element (COH3D16):

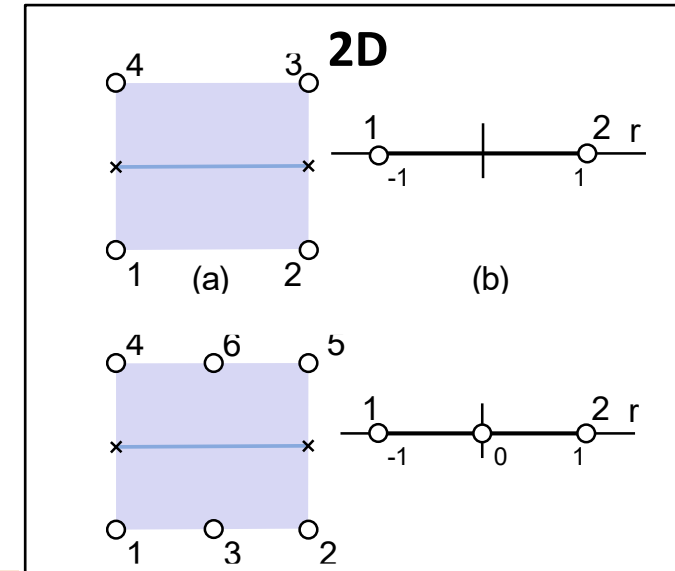
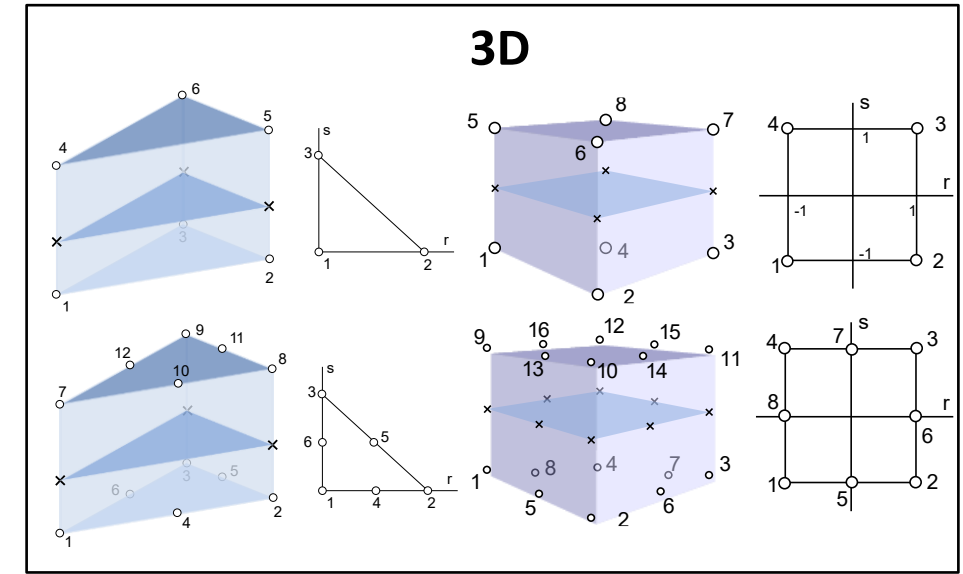


Force transformation:

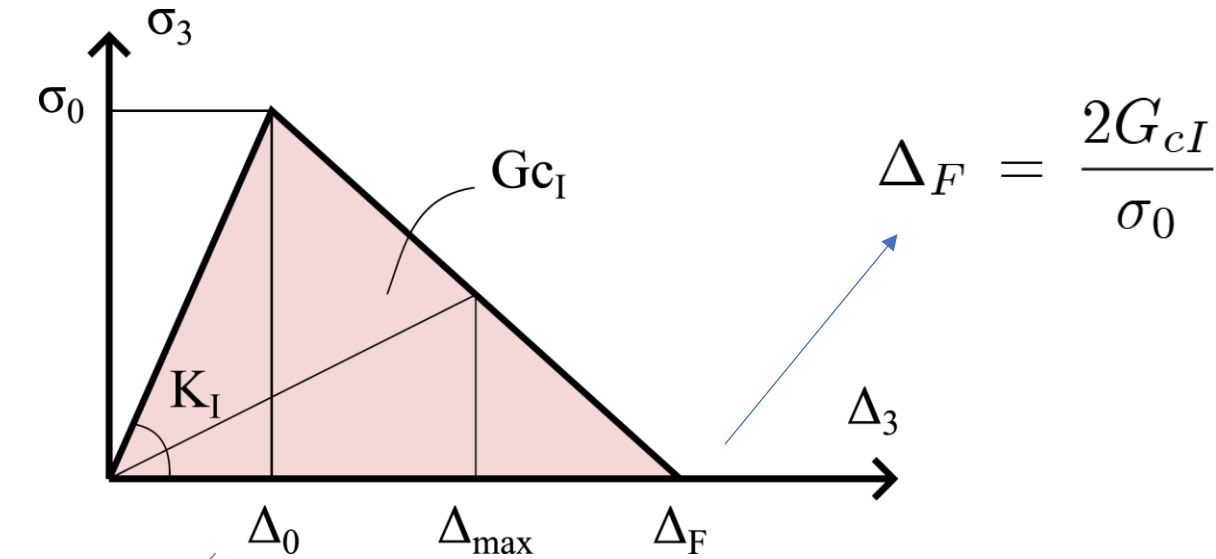
$$\mathbf{f} = \int_{-1}^1 \int_{-1}^1 \Theta^T \boldsymbol{\tau} \|\mathbf{v}_\xi \times \mathbf{v}_\eta\| d\xi d\eta$$

Stiffness transformation:

$$\mathbf{k} = \int_{-1}^1 \int_{-1}^1 \Theta^T \mathbf{D}_{tan} \Theta \|\mathbf{v}_\xi \times \mathbf{v}_\eta\| d\xi d\eta$$



Bi-linear Cohesive Law



Inputs:

Damage initiation strength: σ_0

Penalty stiffness: K_I

Fracture toughness: G_{cI}

$$\Delta_0 = \frac{\sigma_0}{K_I}$$

$$\Delta_{\max} = \Delta_0 + D(\Delta_F - \Delta_0)$$

One state variable (per integration point): Damage D

UEL and .INP file modification

“addcohesive” modifies a polycrystal model

```

SUBROUTINE UEL(RHS,AMATRX,SVARS,ENERGY,NDOFEL,NRHS,NSVARS,
1  PROPS,NPROPS,COORDS,MCRD,NNODE,U,DU,V,A,JTYPE,TIME,DTIME,
2  KSTEP,KINC,JELEM,PARAMS,NDLOAD,JDLTYP,ADLMAG,PREDEF,NPREDF,
3  LFLAGS,MLVARX,DDLMAG,MDLOAD,PNEWDT,JPROPS,NJPROP,PERIOD)
C
C      INCLUDE 'ABA_PARAM.INC'
C
C      DIMENSION RHS (MLVARX,*),AMATRX (NDOFEL,NDOFEL),PROPS (*),
1  SVARS (*),ENERGY (8),COORDS (MCRD,NNODE),U (NDOFEL),
2  DU (MLVARX,*),V (NDOFEL),A (NDOFEL),TIME (2),PARAMS (*),
3  JDLTYP (MDLOAD,*),ADLMAG (MDLOAD,*),DDLMAG (MDLOAD,*),
4  PREDEF (2,NPREDF,NNODE),LFLAGS (*),JPROPS (*)
    
```

of nodes
of inputs (PROPS)
of dimensions (3D)
of integration points

```

*User element, nodes=8, type=U1, properties=9, coordinates=3, variables=4
1, 2, 3
**
*Element, type=U1, elset=UEL
1001, 134, 13, 12, 133, 1332, 1333, 1334, 1335
    
```

User element number
“Top” elements
“Bottom” elements

e.g. COH3D8

```

*Uel Property, elset=UEL
1., 1.e+6, 1.e+6, 100., 100., 5., 5., 2., 0.
    
```

Mode

1: opening only

2: shearing only

3: mixed

Penalty stiffness
Opening: KI Shear: KII

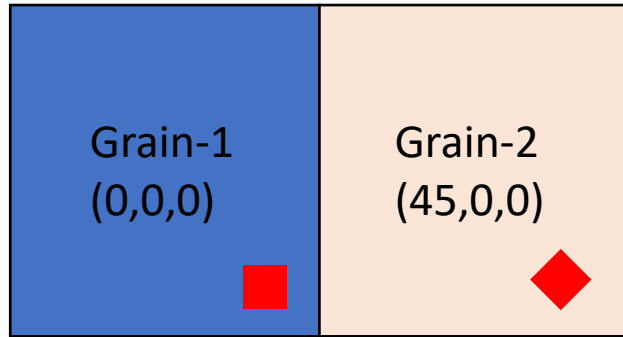
Stress at failure
Opening: SI Shear: SII

Fracture toughness
Opening: GcI Shear: GcII

Height of 2D elements

B-K exponent for mixed mode

DEMO Example



Solid material properties (CP)

Grains with unit dimensions ($1\text{ }\mu\text{m}$)

Uniaxial load
10% strain, 10 s

FCC material – user material-2 (copper)

Mesh: 1-element per grain, C3D8 solid elements

Cohesive model properties

Penalty stiffness: $10^7\text{ MPa}/\mu\text{m}$

Damage initiation strength: 50 MPa

Fracture toughness: $2.5\text{ MPa}\cdot\mu\text{m}$

Mixed mode fracture

Benzeggagh-Kenane exponent: 2

Use the following inputs (“inputs.m”) for “addcohesive” function:

- Section name: “Grain-”
- Material name: “Material”